Cognitive Modeling

Complete Lecture Notes

Week 9: Bayesian Modeling I: Priors, Posteriors and Grid Approximation

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Chapter 1

Bayesian Modeling

1.1 Week 9: Bayesian Modeling I: Priors, Posteriors and Grid Approximation

Chapter 4: Bayesian modeling · Lecture 1 h · Lab 2 h Running data: a recall proportion (30 of 50) and the Week 2 SDT participant (68/100 hits, 11/100 false alarms).

1.1.1 Key terms

Important terms of this lesson, with their Persian equivalents (repeated even if they appeared in earlier weeks).

English	فارسی	توضیح کوتاه
prior	توزیع پیشین (prior)	باور درباره‌ی پارامتر پیش از دیدن داده
posterior	توزیع پسین (posterior)	باور درباره‌ی پارامتر بعد از دیدن داده
grid approximation	تقریب شبکه‌ای (grid)	محاسبه‌ی پسین روی شبکه‌ای از مقادیر
signal detection theory (SDT)	نظریه‌ی تشخیص علامت	مدلی که حساسیت را از سوگیری پاسخ جدا می‌کند
hit	اصابت (hit)	پاسخ «بله» وقتی علامت وجود دارد
Bayes' rule	قاعده‌ی بیز	پیشین × درست‌نمایی ∝ پسین
parameter	پارامتر	کمیت قابل تنظیم مدل که بین افراد یا شرایط فرق می‌کند
likelihood	درست‌نمایی	احتمال داده به‌عنوان تابعی از پارامترها
evidence	شواهد	اطلاعاتی که برای تصمیم جمع می‌شود
beta-binomial model	مدل beta-binomial	با داده‌ی دوجمله‌ای Beta پیشین

English	فارسی	توضیح کوتاه
binomial distribution	توزیع دوجمله‌ای	توزیع تعداد موفقیت در n کوشش
MAP estimate	برآورد MAP	مُد توزیع پسین
credible interval / HDI	بازه‌ی باورپذیر (credible interval)	بازه‌ای که با احتمال مشخص پارامتر را دارد
confidence interval (CI)	بازه‌ی اطمینان	بازه‌ای که با روش مشخص پارامتر واقعی را در بر می‌گیرد
prior predictive check	بررسی پیش‌بینی پیشین	شبیه‌سازی داده از پیشین برای سنجش معقول بودن آن
simulation	شبیه‌سازی	تولید داده‌ی مصنوعی با اجرای مدل
sensitivity (d′)	حساسیت	فاصله‌ی توزیع علامت و نویز
estimate / estimator	برآوردگر / برآورد	روش به / مقدار به‌دست‌آمده برای پارامتر دست آوردن آن
maximum likelihood estimate (MLE)	برآورد بیشینه‌ی درست‌نمایی	پارامترهایی که داده را محتمل‌ترین می‌کنند
cognitive model	مدل شناختی	توصیف دقیق و ریاضی یا محاسباتی از یک فرایند ذهنی
trial	کوشش (trial)	یک بار انجام تکلیف توسط شرکت‌کننده
conjugate prior	پیشین مزدوج (conjugate)	پیشینی که پسین هم‌خانواده می‌دهد
forgetting curve	منحنی فراموشی	کاهش یادآوری با گذشت زمان
accuracy	دقت	نسبت پاسخ‌های درست

1.1.2 Learning outcomes

By the end of this week, students will be able to:

1. State Bayes' rule for parameters and identify the roles of prior, likelihood, evidence and posterior.
2. Compute posteriors analytically for the beta-binomial model and numerically by grid approximation, for one and two parameters.
3. Summarize a posterior (mean, median, MAP, SD, equal-tailed and highest-density intervals) and distinguish a credible interval from a confidence interval.
4. Explain how the prior and the amount of data jointly determine the posterior, and use prior predictive simulation to check whether a prior is sensible.
5. Interpret posterior probabilities of psychologically meaningful hypotheses (e.g., "$P(d' > 1 \mid \text{data})$").

1.1.3 Motivation & big picture

Since Week 3 we have asked: "which single parameter value best accounts for the data?" and then, in Week 5, "how much would that estimate vary across imaginary replications?" Researchers often want to say something more direct:

> *"Given these data, there is a 95% probability that this participant's sensitivity lies between 1.2 and 2.1." "The probability that the payoff manipulation made participants more liberal is 0.98."*

Frequentist confidence intervals do not license such statements (Week 5). Bayesian inference does, at a price: we must state what we believed about the parameters before seeing the data, in the form of a prior distribution. Bayes' rule then combines the prior with the likelihood (which we already know how to write) into a posterior distribution: everything the model and the data together say about the parameters.

Bayesian methods also make several of our earlier headaches easier: uncertainty comes for free (no Hessians or bootstraps), hierarchical models of many participants become natural (Week 13), and model comparison gets a principled answer (Week 10). The cost is computation, which is why Weeks 11–13 are about sampling.

> Reminder (Week 1): this week is about Bayesian data analysis, a method we as scientists use to estimate the parameters of *any* model. It makes no claim that the mind is Bayesian. Bayesian *models of cognition* come in Week 15.

Where is this used?

- Small samples and individual cases. Clinical neuropsychology often has one patient and a few dozen trials. A Bayesian analysis with a sensible prior gives usable, honest uncertainty where MLE is unstable.
- Direct probability statements that clinicians and policy makers understand: "there is a 97% probability that the new training improved memory".
- Adaptive testing: computerized tests (e.g. vision and hearing tests, adaptive exams) update a posterior after every response and choose the next item where it is most informative.
- Combining evidence: today's posterior is tomorrow's prior, so results of earlier studies can be built in formally.
- Modern cognitive modeling software (PyMC, Stan, JAGS) is Bayesian; most hierarchical models in recent papers are fitted this way (Week 13).

1.1.4 Lecture notes

In plain words So far we found *one* best parameter value and then asked how much it might vary. The Bayesian approach instead describes our knowledge about the parameter as a whole probability distribution. Before the data, the prior says which values we consider plausible (e.g. "recall probability is probably moderate, not 0 or 1"). The data then re-weight every possible value by its likelihood (Week 4). The result, the posterior, says how plausible each value is *after* seeing the data. In words: *posterior* $\propto$ *likelihood* $\times$ *prior*. From the posterior we can read off a best guess (its mean), an uncertainty (its spread), and direct statements such as "the probability that $d' > 1$ is 0.93". With little data the prior matters a lot; with much data the posterior is dominated by the likelihood and ends up close to the MLE. Computing the posterior is easy for one or two parameters (on a grid) and needs sampling methods beyond that (Weeks 11–13).

Suggested timing (60 min): Bayes' rule 10 · priors 10 · beta-binomial 12 · grid approximation 10 · summaries and credible intervals 12 · prior predictive checks 6.

1. Bayes' rule for parameters

$$\underbrace{p(\boldsymbol{\theta} \mid \mathrm{y})}_{\text{posterior}} = \frac{\overbrace{p(\mathrm{y} \mid \boldsymbol{\theta})}^{\text{likelihood}}\ \overbrace{p(\boldsymbol{\theta})}^{\text{prior}}}{\underbrace{p(\mathrm{y})}_{\text{evidence (marginal likelihood)}}}, \qquad p(\mathrm{y}) = \int p(\mathrm{y} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta})\, d\boldsymbol{\theta}.$$

(We suppress "$\mathcal{M}$" in the notation this week: everything is conditional on the model.)

- The likelihood is exactly the function from Week 4.
- The prior expresses what is known or plausible before the data.
- The evidence $p(\mathrm{y})$ does not depend on $\boldsymbol{\theta}$. For estimation it is just the normalizing constant that makes the posterior integrate to 1:

$$p(\boldsymbol{\theta} \mid \mathrm{y}) \propto p(\mathrm{y} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta}).$$

Posterior $\propto$ likelihood $\times$ prior. In Week 10 the evidence becomes the star of Bayesian model comparison.

Intuition. The posterior is a compromise: it is high where the parameter values are both plausible *a priori* and make the data probable. With little data the prior matters; with a lot of data the likelihood dominates and the posterior concentrates around the MLE.

Reading the math

- $p(\boldsymbol{\theta} \mid \mathrm{y})$: "probability (density) of the parameter value $\boldsymbol{\theta}$ *given* the data". This is what we want.
- $p(\mathrm{y} \mid \boldsymbol{\theta})$: the likelihood, read as a function of $\boldsymbol{\theta}$ (Week 4).
- $p(\boldsymbol{\theta})$: the prior, chosen before the data.

- $p(\mathrm{y}) = \int p(\mathrm{y} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta})\, d\boldsymbol{\theta}$: the integral adds up "likelihood × prior" over *all* possible parameter values. It is one number, the same for every $\boldsymbol{\theta}$, so for estimation it only rescales the curve to total area 1.
- $\propto$ means "proportional to": equal up to a constant factor. In practice we compute likelihood × prior at many values and then divide by their sum, exactly as in the 4-point table of the worked example (0.0004479 / 0.0009574 = 0.468).

2. Choosing priors

Kind	Example for a recall probability θ	When
Flat / uniform	Beta$(1, 1)$	When every value in the range is genuinely equally plausible (rarely true)
Weakly informative	Beta$(2, 2)$: mild preference for moderate values	Default for most cognitive models: rules out absurd values, otherwise lets the data speak
Informative	Beta$(20, 5)$: "recall after one day is around 0.8"	When previous studies or theory provide real knowledge

Principles: - Priors must be fixed before seeing the data. - They should respect the parameter's range (e.g., no prior mass on negative decay rates). - They should be checked by prior predictive simulation (§6): which *data* does the prior imply? - A "vague" prior on one scale can be very informative on another scale (lab Part D). - Report your priors and check the sensitivity of conclusions to reasonable alternatives (Week 10).

3. A closed-form case: the beta-binomial model

For y successes in n binomial trials with a Beta(a_0, b_0) prior on θ:

$$p(\theta \mid y) \propto \underbrace{\theta^{y}(1-\theta)^{n-y}}_{\text{likelihood}} \cdot \underbrace{\theta^{a_0-1}(1-\theta)^{b_0-1}}_{\text{prior}} = \theta^{a_0+y-1}(1-\theta)^{b_0+n-y-1},$$

so the posterior is again a Beta distribution:

$$\theta \mid y \sim \text{Beta}(a_0 + y,\ b_0 + n - y).$$

When prior and posterior come from the same family, the prior is called conjugate.

- a_0 and b_0 act like pseudo-counts of prior successes and failures.

- Posterior mean: $\dfrac{a_0 + y}{a_0 + b_0 + n}$, a weighted average of the prior mean $\frac{a_0}{a_0+b_0}$ and the observed proportion y/n, with weights $a_0 + b_0$ and n.
- Connection to Weeks 2 and 4: the log-linear correction $(y+0.5)/(n+1)$ is exactly the posterior mean under a Beta$(0.5, 0.5)$ prior, the so-called *Jeffreys prior*. The "correction" was a Bayesian estimate in disguise.

Reading the math

- The Beta distribution Beta(a, b) is a flexible bell-like curve on the interval 0–1, ideal for describing beliefs about a probability. Its mean is $a/(a + b)$. Beta$(1, 1)$ is flat; Beta$(20, 5)$ is a narrow bump around 0.8.
- The update rule Beta$(a_0, b_0) \to$ Beta$(a_0 + y, b_0 + n - y)$ is just counting: add the observed successes to a and the failures to b. The prior behaves as if we had already seen a_0 successes and b_0 failures (pseudo-counts).
- Example: prior Beta$(2, 2)$ ("as if" 2 recalls and 2 failures), data 7 of 10 $\to$ Beta$(9, 5)$, mean $9/14 = 0.643$. The prior pulls the observed 0.70 slightly toward 0.5, because it adds 4 pseudo-observations to 10 real ones.

4. Grid approximation

Most cognitive models have no conjugate prior. If there are only one or two parameters, we can approximate the posterior on a grid:

1. Define a grid of parameter values $\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_G$ covering all plausible values.
2. At each grid point compute the log prior plus the log likelihood (logs avoid underflow, as in Week 4).
3. Subtract the maximum, exponentiate, and normalize so that the values sum to 1 (or integrate to 1).

The result is a discrete approximation of the posterior from which we can compute any summary. Grid approximation is exact up to the grid resolution and is ideal for building intuition. It fails beyond 2–3 parameters (the curse of dimensionality, Week 3), which motivates MCMC (Weeks 11–13).

5. Summarizing a posterior

Summary	Definition	Remark
Posterior mean	$\mathbb{E}[\theta \mid \mathrm{y}]$	Minimizes expected squared error
Posterior median	50th percentile	Robust to skew
MAP	$\arg\max_\theta p(\theta \mid \mathrm{y})$	Equals the MLE under a flat prior

Summary	Definition	Remark
Posterior SD	$\sqrt{\mathrm{Var}[\theta \mid \mathrm{y}]}$	The Bayesian counterpart of a SE
95% equal-tailed interval	2.5th to 97.5th percentiles	Easy to compute
95% HDI (highest-density interval)	The narrowest interval containing 95% of the posterior; every value inside is more probable than every value outside	Preferred for skewed posteriors

Credible vs. confidence interval. A 95% credible interval *does* mean "given the model, the prior and the data, the parameter lies in this interval with probability 0.95". A confidence interval is a statement about the long-run behavior of the procedure (Week 5). With weak priors and plenty of data, the two often coincide numerically, but their interpretations differ.

Posterior probabilities of hypotheses come for free: $P(d' > 1 \mid \mathrm{y})$ is just the posterior mass above 1.

6. Prior predictive checks

The prior predictive distribution describes the data the model expects *before* seeing any data:

$$p(\mathrm{y}^{\mathrm{rep}}) = \int p(\mathrm{y}^{\mathrm{rep}} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta})\, d\boldsymbol{\theta}.$$

To simulate from it: draw $\boldsymbol{\theta}$ from the prior, then simulate data from the model. If the simulated data look absurd (e.g., nearly all hit rates at exactly 0 or 1, or recall that *increases* over a month), the prior is unreasonable. Prior predictive checks are the Bayesian version of the "simulate before you fit" habit from Week 2.

> Advanced: the posterior as updated prior. Because $p(\boldsymbol{\theta} \mid \mathrm{y}_1, \mathrm{y}_2) \propto p(\mathrm{y}_2 \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta} \mid \mathrm{y}_1)$ for independent data, today's posterior is tomorrow's prior. Bayesian updating is therefore naturally sequential, which is also why Bayesian inference is a popular normative model of *learning* (Week 15).

For an engineer

- This is Bayesian estimation as in detection and estimation theory: the posterior is what an MAP or MMSE estimator works with. The posterior mean is the MMSE estimate; the posterior mode is the MAP estimate; with a flat prior MAP = ML.
- The conjugate Beta–binomial update is the same idea as conjugate Gaussian updating, the scalar core of the Kalman filter (prior mean and variance + measurement → posterior mean and variance), here for a probability instead of a Gaussian state.

- Sequential updating ("today's posterior is tomorrow's prior") is exactly recursive Bayesian filtering.
- What is different in psychology: the prior is usually weak and must be *justified to readers*, and the main output is not a decision but a description of uncertainty about a psychological parameter. Prior predictive checks are the habit that makes priors defensible.

For the instructor

- Do the 4-point grid on the board (worked example, Part 1) with a column for prior, likelihood, product and posterior. It contains the whole of Bayesian inference in one table. Then change the prior to (0.1, 0.1, 0.4, 0.4) and recompute live: students see the prior at work.
- Use the Beta pseudo-count story. "A Beta(2, 2) prior is like having already seen 2 recalls and 2 failures." Students who find priors arbitrary relax once they see them as imaginary data.
- Address the "subjectivity" worry directly. Every analysis has assumptions (the likelihood is one too); Bayesian analysis makes one more of them explicit, and sensitivity analysis (Week 10) checks how much it matters.
- Typical questions.
  - *"Which prior is correct?"* None is uniquely correct. Use weakly informative priors, check them with prior predictive simulations, and report sensitivity.
  - *"Is a credible interval the same as a confidence interval?"* Often similar numerically with weak priors and enough data, but only the credible interval means "95% probability that the parameter is in here".
  - *"Is the brain Bayesian, then?"* That is a different question (Week 15). This week Bayes is a tool for the scientist.

1.1.5 Worked example

Part 1: a four-point grid by hand

A participant recalls $y = 7$ of $n = 10$ items. To keep the arithmetic simple, suppose θ can only take the values 0.2, 0.4, 0.6 or 0.8, each with prior probability 1/4.

θ	Prior	Likelihood $\theta^7(1-\theta)^3$ (constant dropped)	Prior × likelihood	Posterior
0.2	0.25	0.0000066	0.0000016	0.002
0.4	0.25	0.0003539	0.0000885	0.092
0.6	0.25	0.0017916	0.0004479	0.468
0.8	0.25	0.0016777	0.0004194	0.438

θ	Prior	Likelihood $\theta^7(1-\theta)^3$ (constant dropped)	Prior × likelihood	Posterior
Sum	1		0.0009574	1.000

Each posterior value = (prior × likelihood) / sum. The binomial coefficient was dropped because it cancels in the normalization. Interpretation: after the data, 0.6 and 0.8 are about equally plausible, and 0.2 is practically excluded.

Part 2: the same data with a continuous Beta prior

Prior Beta$(2, 2)$ (mean 0.5, mildly favoring moderate values). Posterior: Beta$(2 + 7, 2 + 3) =$ Beta$(9, 5)$.

- Posterior mean $= 9/14 = 0.643$: between the prior mean (0.5) and the observed proportion (0.7), closer to the data because $n = 10 > a_0 + b_0 = 4$.
- MAP (mode) $= (9 - 1)/(9 + 5 - 2) = 0.667$.
- 95% equal-tailed credible interval $= [0.386, 0.861]$.

Interpretation. Given a mild prior and 10 items, the recall probability is probably around 0.64, but anything from about 0.39 to 0.86 remains credible. Ten items tell us little.

Part 3: the prior matters less with more data

With $y = 30$ of $n = 50$:

Prior	Posterior	Mean	95% credible interval
Beta$(1, 1)$ (flat)	Beta$(31, 21)$	0.596	[0.461, 0.724]
Beta$(0.5, 0.5)$ (Jeffreys)	Beta$(30.5, 20.5)$	0.598	[0.462, 0.727]
Beta$(20, 5)$ (informative, "≈ 0.8")	Beta$(50, 25)$	0.667	[0.557, 0.768]

The flat and Jeffreys priors give practically the same answer, also equal to the Week 5 profile-likelihood interval [0.462, 0.728]. The informative prior, which is worth 25 pseudo-observations, still pulls the estimate noticeably toward 0.8: 50 observations are not enough to overwhelm a strong prior. Whether that is good depends on whether the prior knowledge was right.

1.1.6 Lab (2 hours)

Goals

1. Compute beta-binomial posteriors analytically and by grid approximation, and see the prior–data compromise.
2. Watch the posterior sharpen with data (sequential updating).
3. Compute a 2-D grid posterior for SDT (d', c); summarize it with means, HDIs and posterior probabilities.
4. Use prior predictive simulation to expose a bad "vague" prior.

Timing: Part A 25 min · Part B 15 min · Part C 40 min · Part D 25 min · Part E 15 min.

Setup

Code · کد: `code/week09.py`, cell 1 · سلول 1

Part A: beta-binomial, analytically and on a grid (25 min)

Code · کد: `code/week09.py`, cell 2 · سلول 2

Expected output:

```
flat Beta(1,1)           grid mean = 0.5962, exact mean = 0.5962, 95% CI
↪  = [0.461 0.724]
Jeffreys Beta(.5,.5)     grid mean = 0.5980, exact mean = 0.5980, 95% CI
↪  = [0.462 0.727]
informative Beta(20,5)   grid mean = 0.6667, exact mean = 0.6667, 95% CI
↪  = [0.557 0.768]
```

The grid and exact posteriors coincide. Look at where each posterior sits relative to its prior and the likelihood.

Part B: sequential updating (15 min)

Code · کد: `code/week09.py`, cell 3 · سلول 3

What to notice: the posterior starts as the prior, narrows as items accumulate, and concentrates around the true value. Its width shrinks roughly like $1/\sqrt{n}$, as the SE did in Week 5.

Part C: a two-parameter grid posterior for SDT (40 min)

The Week 2 participant: 68 hits out of 100 signal trials and 11 false alarms out of 100 noise trials. Priors: $d' \sim \mathcal{N}(1, 1.5)$ (sensitivity is usually positive, rarely above 4) and $c \sim \mathcal{N}(0, 1)$ (bias is usually modest).

> Code · کد: `code/week09.py`, cell 4 · سلول 4

Expected output:

```
    post. mean  post. SD    MAP  95% HDI low  95% HDI high
d'       1.688     0.209  1.675        1.250         2.062
c        0.376     0.106  0.375        0.163         0.575
MLE (uncorrected formulas): d' = 1.694, c = 0.379
P(d' > 1 | data) = 1.000;  P(c > 0 | data) = 1.000
Posterior correlation of d' and c: 0.245
```

("1.000" means that none of the 20 000 posterior samples fell below the threshold, so the probability is > 0.9999. Report it as such, not as certainty.)

Interpret (write in your notebook): 1. How do the posterior mean and MAP compare with the MLE? Why are they so close here? 2. Translate the HDI for d' and $P(c > 0 \mid \text{data})$ into one plain-language sentence each about this participant. 3. Increase the prior SD on d' to 10, then decrease it to 0.3 (centered at 1). Which summaries change, and by how much?

Part D: prior predictive checks (25 min)

Before trusting a prior, look at the data it implies. Compare our prior with a "vague" one, $d' \sim \mathcal{N}(0, 10)$ and $c \sim \mathcal{N}(0, 10)$, which is sometimes used because it is "uninformative".

> Code · کد: `code/week09.py`, cell 5 · سلول 5

Expected output:

```
our prior: d'~N(1,1.5), c~N(0,1): 14% of simulated datasets have a rate
↪  of exactly 0 or 1; 24% have H < F
'vague' prior: d'~N(0,10), c~N(0,10): 96% of simulated datasets have a
↪  rate of exactly 0 or 1; 23% have H < F
```

What to notice: the "vague" prior is not neutral at all. On the probability scale it puts 96% of simulated datasets at hit or false-alarm rates of exactly 0 or 1, which practically never happens in real detection experiments. It expresses a strong and absurd belief about the data. Vagueness on the parameter scale is not vagueness on the data scale.

Our own prior is not perfect either: about a quarter of its simulated datasets have $H < F$, because $\mathcal{N}(1, 1.5)$ puts about 25% of its mass on negative d'. For a task where participants are known to be

above chance, a prior restricted to $d' > 0$ (e.g., a half-normal or a Gamma) would be more defensible. Try it, and check whether it changes the posterior for this participant. (Hint: with 200 informative trials, hardly at all.)

Part E: interpret (15 min)

Write a short "results paragraph" (4–6 sentences) for this participant as it would appear in a paper. Report the priors, the posterior means and 95% HDIs for d' and c, $P(d' > 1 \mid \text{data})$, and one sentence on prior sensitivity (from Part C, question 3).

Exercises (increasing difficulty)

Exercise 1 $\star$. Show algebraically that the posterior mean of the beta-binomial model is a weighted average of the prior mean and y/n. What are the weights?

Exercise 2 $\star$. Using the Part A grid, compute the 95% HDI and the equal-tailed interval for Beta$(1 + 2, 1 + 18)$ (2 successes out of 20). Why do they differ?

Exercise 3 $\star\star$ (forgetting by grid). Compute the grid posterior of (A, d) for the power forgetting model with the Week 2 data, using priors $A \sim \text{Beta}(8, 2)$ and $d \sim \text{Gamma}(\text{shape } 2, \text{scale } 0.3)$. Compare the posterior means and 95% HDIs with the MLE and the Week 5 profile and bootstrap intervals.

Exercise 4 $\star\star$ (equivalence of MAP and MLE). In Part C, replace both priors by flat priors over the grid. Show that the MAP equals the MLE. Why is the posterior *mean* still slightly different?

Exercise 5 $\star\star\star$ (Advanced: sequential vs. batch). For the SDT model, update the grid posterior trial by trial (each trial's posterior becomes the next trial's prior) through 200 simulated trials, and show that the final posterior is identical (up to numerical error) to the batch posterior from all trials at once. Plot the posterior SD of d' against the number of trials.

Lab check-off

Show: (1) the three-panel prior/likelihood/posterior plot from Part A, (2) the SDT summary table and joint posterior from Part C, and (3) the two prior predictive plots from Part D with one sentence on what is wrong with the vague prior.

1.1.7 Connection to AI & LLMs

How good is the model, really? A Bayesian answer for small evaluations. Researchers often test an LLM on a small set of items, for example 25 hand-written problems. The beta-binomial model of this week gives what such studies usually lack: a full posterior for the accuracy, direct probability statements ("the probability that accuracy exceeds 60%"), and a direct comparison of two models.

Code · کد: `code/week09.py`, cell 6 · سلول 6

Expected output:

```
Model A: posterior mean = 0.704, 95% interval = [0.522, 0.857],
 ↪  P(accuracy > 0.6) = 0.878
Model B: posterior mean = 0.593, 95% interval = [0.406, 0.766],
 ↪  P(accuracy > 0.6) = 0.479
P(A more accurate than B | data) = 0.807
```

With 25 items, the posterior intervals are about 0.35 wide, and there is a 19% posterior probability that B is actually the better model.

Try it (no API needed). - ⋆ Use an informative prior from an earlier, larger evaluation (e.g., Beta(14, 6) for model A). How much do the conclusions change? - ⋆⋆ The comparison above treats the two models as independent, although they answered the same items. Why is this a problem (see the Week 5 box)? Sketch a model that respects the pairing. - ⋆⋆ Discussion: what prior would you use for the accuracy of a brand-new model on a four-choice test, and why should it not put much mass below 0.25?

1.1.8 Common mistakes & misconceptions

1. "The prior makes Bayesian inference subjective, so the results are arbitrary." Priors are explicit assumptions, just like the choice of likelihood. Report them, justify them, and check sensitivity. With adequate data, reasonable priors lead to similar conclusions.
2. "Flat or very wide priors are uninformative." They can be strongly informative on the data scale (Part D). Use weakly informative priors and check them with prior predictive simulation.
3. Choosing the prior after looking at the data. That uses the data twice.
4. Multiplying probabilities on a grid without logs, which causes underflow (Week 4). Work with log-prior + log-likelihood, then subtract the maximum before exponentiating.
5. Too coarse or too narrow a grid. If the posterior is cut off at the grid edge, all summaries are wrong. Check that the posterior is near zero at the edges.
6. Reading a credible interval as a confidence interval, or vice versa. Only the credible interval supports "95% probability the parameter is in here" (given the model and prior).
7. Reporting only the MAP. The point of the posterior is the whole distribution. Report a central value *and* an interval, and probabilities for the hypotheses you care about.

1.1.9 Check your understanding

1. Write Bayes' rule for a parameter θ and name each term. Answer $p(\theta \mid y) = p(y \mid \theta)p(\theta)/p(y)$: posterior = likelihood × prior / evidence (marginal likelihood). For estimation, $p(\theta \mid y) \propto p(y \mid \theta)p(\theta)$.

2. Prior Beta$(3, 3)$, data: 12 recalled out of 20. What is the posterior, and its mean? Answer Beta$(3 + 12, 3 + 8) =$ Beta$(15, 11)$, with mean $15/26 = 0.577$.

3. Why does the posterior get closer to the MLE as n increases? Answer The log-likelihood is a sum over observations and grows with n, while the log-prior stays fixed. With much data the likelihood dominates the product, and the posterior concentrates around the likelihood's maximum.

4. What is the difference between a 95% HDI and a 95% equal-tailed interval? When do they coincide? Answer The HDI is the narrowest interval containing 95% of the posterior mass, and every value inside has higher density than any value outside. The equal-tailed interval cuts 2.5% from each tail. They coincide for symmetric unimodal posteriors and differ for skewed ones.

5. Why does grid approximation become impractical for models with many parameters? Answer The number of grid points grows exponentially with the number of parameters (e.g., 100 points per parameter means 100^k evaluations). That is the curse of dimensionality. MCMC sampling (Weeks 11–13) avoids it.

6. A colleague uses $d' \sim \mathcal{N}(0, 100)$ "to be safe". What would you do to convince them it is a bad idea? Answer A prior predictive simulation: show that this prior implies hit and false-alarm rates that are almost always exactly 0 or 1, an absurd belief about real detection data. Suggest a weakly informative prior instead.

7. Explain the statement "the log-linear correction for extreme rates is a Bayesian estimate". Answer The corrected rate $(y + 0.5)/(n + 1)$ equals the posterior mean of θ under a Beta$(0.5, 0.5)$ prior: $(0.5 + y)/(0.5 + 0.5 + n)$. The correction implicitly adds half a pseudo-observation of each type.

1.1.10 Assignment

- A3 (model comparison and recovery) and the project proposal are due at the start of this week's lecture.
- No new graded assignment this week. A4 (Bayesian analysis by grid) is released in Week 10.
- Optional practice (ungraded): lab Exercise 3 (forgetting by grid) is the core of A4. Starting it now will save you time.

1.1.11 Further reading

- Kruschke (2014), chapters 5–6. Bayes' rule, and inferring a binomial probability via exact mathematical analysis (the beta-binomial model). Very gentle and visual. *(Main reading.)*
- Lee & Wagenmakers (2014), chapters 1 and 3. The basics of Bayesian analysis, and inferences with binomials.
- Martin, Kumar & Lao (2021), chapter 1. Bayesian inference: the components of a Bayesian model and prior/posterior predictive distributions, with Python code.

- Farrell & Lewandowsky (2018), chapter 6. Bayesian parameter estimation: basic concepts, linked to the likelihood framework of Week 4.
- Azimifar & Ahmadi (1396), vol. 1 (in Persian): the section on Bayesian parameter estimation and its relation to ML estimation [verify chapter].

Preview for Week 10: the posterior tells us about parameters *within* a model. Next week we ask how the data change our belief *between* models, using the evidence $p(\mathrm{y} \mid \mathcal{M})$ that we have been ignoring as a normalizing constant, and how to check whether a model's posterior predictions look like the real data.